LM Physics of Climate

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5 Radiative transfer

Radiation is a very important form of energy transport in planetary atmospheres, and typically dominates in the upper atmosphere. In this section, we will be deriving how radiation interacts with the gas in an atmosphere.

5.1 Optical depth

As radiation passes through a gas, some of it is blocked by the gas molecules. But how much? On Earth, we are able to see visible light from the Sun, so we know it isn't being totally blocked by the atmosphere. On Venus, however, we can't see through the atmosphere to the surface, so visible light is fully blocked by the atmosphere.

'Optical depth' is a measure of how opaque a substance is to radiation. It can be thought of as the fraction of radiation which is blocked by particles in the gas. When a photon interacts with a molecule, it can be scattered or absorbed and re-emitted (not necessarily in the direction of the initial photon). In either case, the photon is removed from the initial beam.

Molecules have an effective cross-sectional area within which they interact with radiation, commonly labelled as σ. This cross section is wavelength-dependent, and different for scattering vs absorption. For now, we will use σ to denote the overall 'extinction' cross section, i.e. absorption and scattering combined.

In Figure 1, radiation passes through a cylinder of volume $A\delta s$ containing N particles. We define the number density of the particles as $n = N/A\delta s$. For a given wavelength, λ, each particle has a cross section σ. The fractional area of the cylinder blocked by the particles is therefore:

$$\begin{aligned}\frac{N\sigma}{A} &= \frac{n\sigma A\delta s}{A} \\ &= n\sigma\delta s.\end{aligned}$$

We define the optical depth of a small cylinder of length $\mathrm{d}s$ to be:

$$\mathrm{d}\tau = \sigma n \mathrm{d}s \tag{5.1}$$

$$= \kappa\rho \mathrm{d}s, \tag{5.2}$$

where ρ is the mass density of the particles, and κ is the cross section per unit mass.

Note that in Earth atmosphere studies, the sign convection typically used is for optical depth to increase with height in the atmosphere. In astrophysics the opposite is true: optical depth increases as you go deeper into the atmosphere from the exterior of the planet. Each convention reflects the type of data used in the respective field, i.e., measurements from the Earth's surface vs measurements of distant exoplanets and stars from Earth. In this course, we will use the Earth atmosphere convention, as shown in equations 5.1 and 5.2.

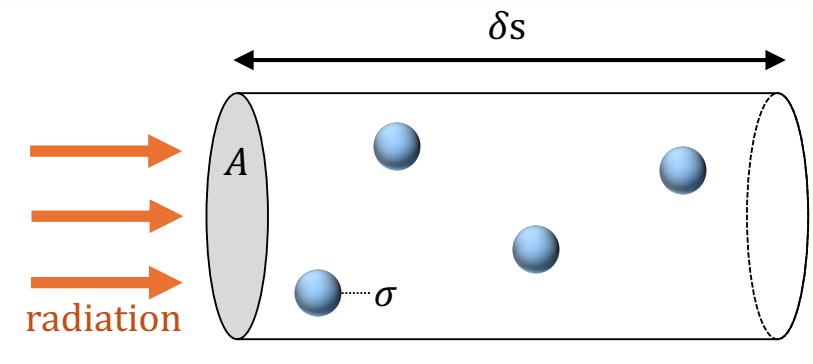


Figure 1: As radiation passes through a cylinder of gas with thickness δs and cross-sectional area A, it is blocked by the particles it encounters, which each have cross-sectional area σ. Optical depth quantifies the fraction of the light which is blocked by the gas.

When $\tau \ll 1$, the gas is almost transparent and we can describe it as being *optically thin*. For example, the Earth's atmosphere is optically thin to visible radiation. When $\tau \approx 1$, the gas becomes opaque and we can describe it as being *optically thick*. Note that $\tau = 1$ corresponds to the case where the particles perfectly cover the entire surface area of the cylinder of gas (Figure 1). It is also possible for $\tau \gg 1$, for example in thick atmospheres such as that of Venus.

5.2 Specific intensity and surface flux

Specific intensity of radiation I_ν is defined as the energy per unit area (A) per unit solid angle (Ω) per unit frequency (ν) per unit time (t):

$$\mathrm{d}I_\nu = \frac{\mathrm{d}E}{\mathrm{d}A\mathrm{d}\Omega\mathrm{d}\nu\mathrm{d}t}. \tag{5.3}$$

Note that I_ν has the same units as the Planck function,

$$B_\nu = \frac{2h\nu^3}{c^2}\frac{1}{\exp\frac{h\nu}{k_\mathrm{B}T} - 1}. \tag{5.4}$$

Both I_ν and B_ν can alternatively be written in terms of wavelength, λ, and are then labelled I_λ and B_λ, respectively. Since the total energy represented by each form must be the same, the frequency vs wavelength forms are related by:

$$I_\nu \mathrm{d}\nu = I_\lambda \mathrm{d}\lambda$$
$$B_\nu \mathrm{d}\nu = B_\lambda \mathrm{d}\lambda$$

Specific intensity is closely related to flux, F_ν (or F_λ), which is defined as the energy per unit time per unit area per unit frequency (or wavelength). The following equations are the same for both the ν and λ-dependent versions of specific intensity and flux, so we will drop the ν and λ subscripts.

The relevant area for the flux is the area perpendicular to the direction of the radiation (for a surface which emits uniformly in all directions, known as Lambert's Cosine Law), so flux can be written in terms of specific intensity:

$$F = \int I\cos(\theta)\mathrm{d}\Omega, \tag{5.5}$$

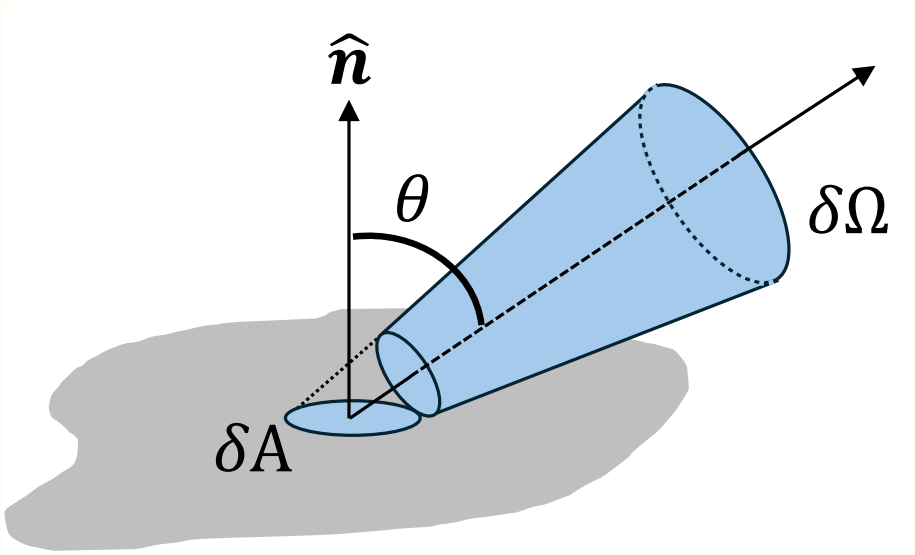


Figure 2: Schematic of radiation emitted from a surface element δA in a solid angle element $\delta\Omega$. The normal to the surface is $\hat{\mathbf{n}}$, which subtends an angle θ to the direction of the radiation.

where θ is the angle between the normal and the direction of the radiation (around which the element of solid angle, $\mathrm{d}\Omega$, in centred, see Figure 2).

Let's consider the flux emitted upwards from a surface (called the 'surface flux'), assuming that I is *isotropic*, i.e. independent of direction. In spherical polar coordinates, we can rewrite the flux integral as:

$$\begin{aligned} F &= \int I\cos(\theta)\mathrm{d}\Omega \\ &= \int_0^{2\pi}\int_0^{\pi/2} I\cos(\theta)\sin(\theta)\mathrm{d}\theta\mathrm{d}\phi \\ &= 2\pi\int_0^{\pi/2} I\cos(\theta)\sin(\theta)\mathrm{d}\theta \\ &= 2\pi I\int_0^{1}\cos(\theta)\mathrm{d}(\cos(\theta)) \\ &= \pi I. \end{aligned} \tag{5.6}$$

For a blackbody radiator, $I = B$, and the surface flux is πB. We can calculate its total luminosity, L, by integrating the surface flux over wavelength and area. Assuming that the specific intensity is also *homogenous*, i.e. uniform across the surface, we have:

$$\begin{aligned} L &= \int\int \pi B_\nu \mathrm{d}\nu\mathrm{d}A \\ &= A\int \pi B_\nu \mathrm{d}\nu \\ &= A\sigma_{SB}T^4, \end{aligned}$$

where σ_{SB} is the Stefan-Boltzmann constant, and T is the temperature of the blackbody radiator.

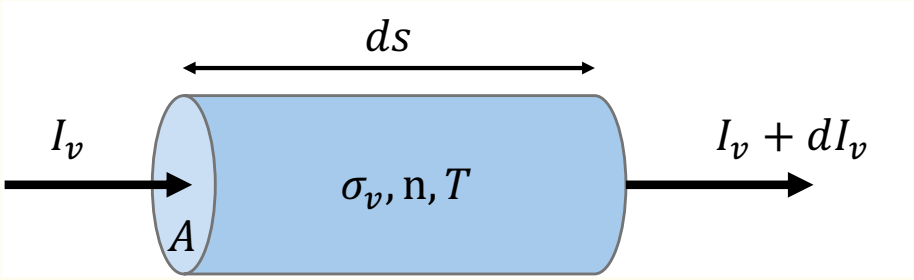


Figure 3: As radiation passes through a thickness ds of gas, it is reduced by absorption and scattering from the gas, and added to by thermal emission from the gas. σ_ν, n and T represent the extinction cross section, number density and temperature of the gas, respectively.

5.3 The radiative transfer equation

To understand an atmosphere and its climate, it is important to know how radiation is transporting energy. When radiation travels in a particular direction through a gas, absorption and scattering from particles (including gas particles, cloud particles etc.) can reduce the amount of radiation travelling in that direction. Equally, the particles can thermally emit radiation, adding to the initial intensity.

The radiative transfer equation states how the specific intensity of radiation changes due to absorption, scattering and thermal emission. Here, we will consider only absorption and thermal emission for simplicity. We will also reintroduce the ν subscript to denote dependence on frequency. Note that the equation of radiative transfer remains the same when dependence on ν is swapped for dependence on λ across all terms.

In Figure 3, a specific intensity I_ν enters a cylinder of gas. Some energy is added to the radiation field due to thermal emission from the gas. The energy added per unit time per unit solid angle per unit frequency is:

$$A\mathrm{d}I_{\nu,emit} = B_\nu(T)\sigma_\nu N = B_\nu(T)\sigma_\nu nA\mathrm{d}s,$$

where T is the temperature of the gas, A is the cross-sectional area of the gas cylinder and N is the number of particles. The factor of σ_ν is present because the gas does not emit as a blackbody - σ_ν tells us how efficiently the gas absorbs or emits radiation at any given wavelength relative to a blackbody. Rearranging slightly, we get:

$$\mathrm{d}I_{\nu,emit} = B_\nu(T)\sigma_\nu n\mathrm{d}s. \tag{5.7}$$

Some radiation is also removed due to absorption by the gas. The energy removed per unit time per unit solid angle per unit frequency is:

$$A\mathrm{d}I_{\nu,abs} = I_\nu(T)\sigma_\nu N = I_\nu\sigma_\nu nA\mathrm{d}s,$$

therefore:

$$\mathrm{d}I_{\nu,abs} = I_\nu\sigma_\nu n\mathrm{d}s.$$

Putting these together, the change in I_ν as it travels across the distance ds of gas is:

$$\begin{aligned}\mathrm{d}I_\nu &= B_\nu(T)\sigma_\nu n\mathrm{d}s - I_\nu\sigma_\nu n\mathrm{d}s \\ \frac{\mathrm{d}I_\nu}{\mathrm{d}s} &= \sigma_\nu n(B_\nu(T) - I_\nu).\end{aligned} \tag{5.8}$$

This is known as the *radiative transfer equation*, and it is used in many contexts, from astrophysics to laboratory studies. When applied to atmospheres, it is useful to

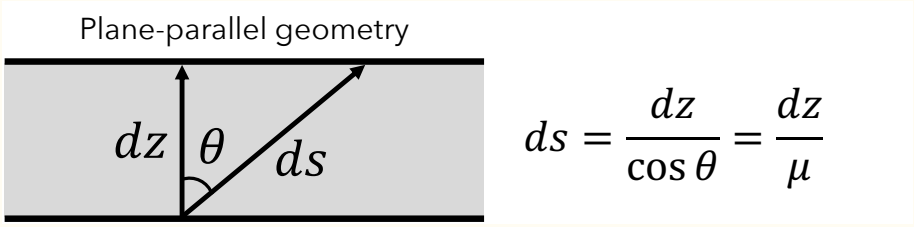


Figure 4: The plane parallel geometry, in which the atmosphere is assumed to be flat.

reframe the radiative transfer equation in the *plane parallel geometry*. In this geometry, we assume that the atmosphere constitutes flat 'slabs' of gas stacked vertically, without accounting for the curvature of the planet (Figure 4). It is a reasonable assumption to make when the atmosphere of a planet is thin relative to its total radius, as is the case for the Earth.

In the plane-parallel geometry, we can rewrite $\mathrm{d}s = \mathrm{d}z/\cos(\theta) = \mathrm{d}z/\mu_\theta$, where θ is the angle between $\mathrm{d}s$ and the vertical, and $\mu_\theta = \cos(\theta)$. Substituting this into the radiative transfer equation (Equation 5.8), we get:

$$\mu_\theta \frac{\mathrm{d}I_\nu}{\mathrm{d}z} = \sigma_\nu n (B_\nu(T) - I_\nu). \tag{5.9}$$

It is also useful to express the radiative transfer equation in terms of the optical depth, τ. Combining equations 5.1 and 5.9, we get:

$$\mu_\theta \frac{\mathrm{d}I_\nu}{\mathrm{d}\tau} = B_\nu(T) - I_\nu. \tag{5.10}$$

5 This equation is central to calculating the radiation field in a planetary atmosphere, and how it shapes the vertical temperature profile.

5.4 Absorption cross sections

From Equation 5.9, we see that the absorption cross sections of the gas, σ_ν, are central to determining the radiative transfer of energy in an atmosphere. Each type of molecule has a unique spectral signature, which can be calculated theoretically and/or measured in the lab (indeed, this is a rich area of research in itself). Figure 5 shows the absorption cross sections for H_2O, CO_2, O_2 and N_2 from 0.4–50 μm at 300 K and 1 bar. H_2O and CO_2 are particularly strong absorbers, which makes them excellent greenhouse gases. O_2 and N_2 have much smaller absorption cross sections and therefore play a more minor role in our atmosphere's radiative transfer.

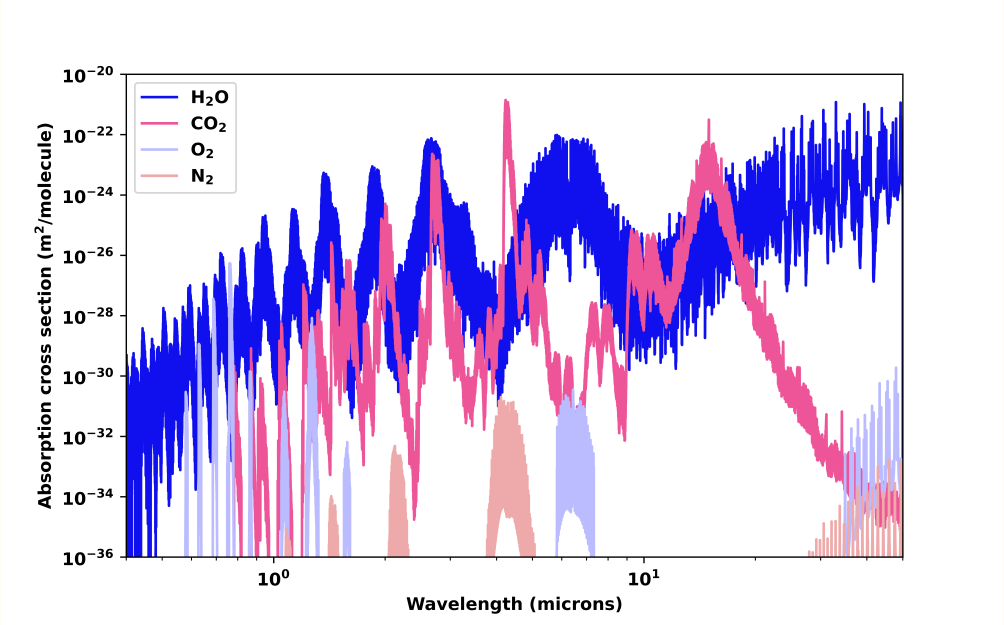


Figure 5: The absorption cross sections for H_2O, CO_2, O_2 and N_2 at 300 K and 1 bar.